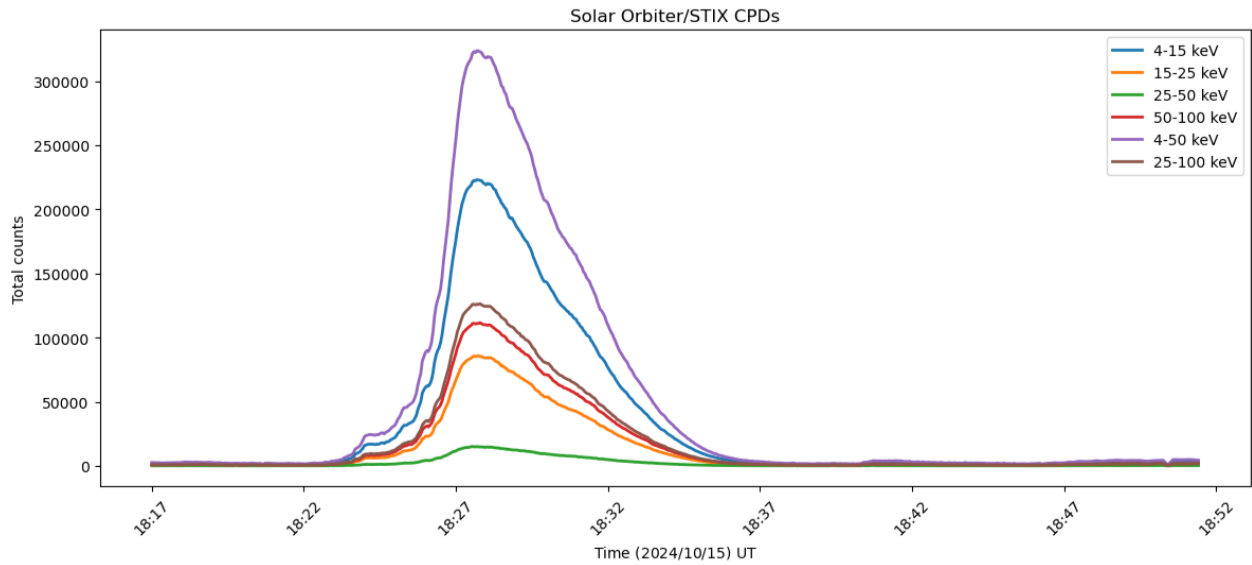


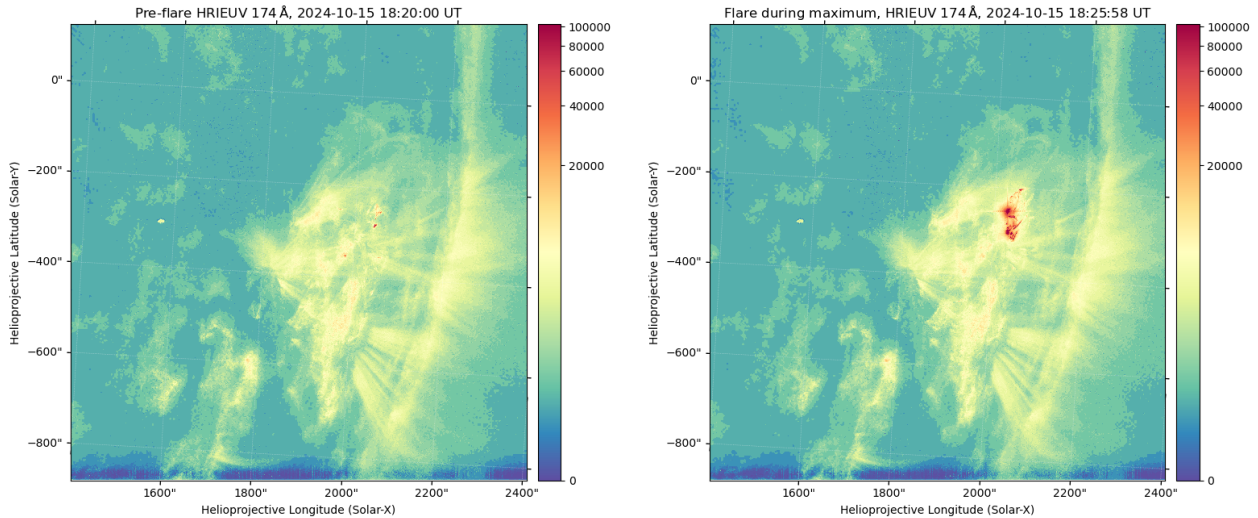
Summer student internship: Multiwavelength Analysis of Quasi-Periodic Pulsations (QPPs) in a Solar Flare | **PROGRESS REPORT no. 2** | 01/08/2025

For the second progress report we analyzed M2.0-class solar flare (15/10/2024 around 18 : 30 UT), which was observed during the Solar Orbiter Major Flare Watch Campaign. Analysis was done through multiple steps, and here we're showing key plots and conclusions.

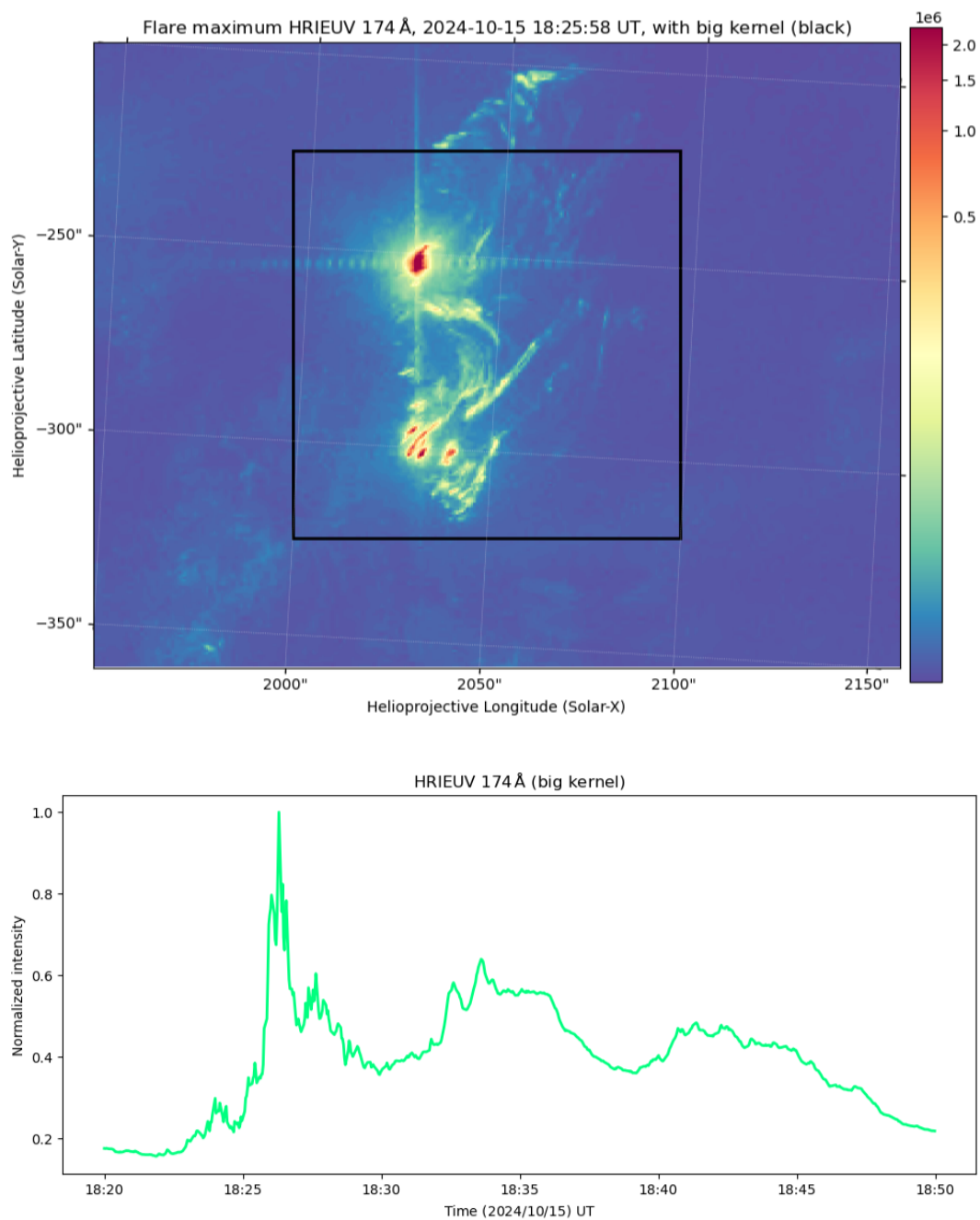
Bellow, we show Compressed Pixel Data (CPD) from STIX - those are more suitable for general examination and detailed QPPs analysis.



Furthermore, we show the observed region ($\lambda = 174 \text{ \AA}$) for pre-flare and flare maximum timestamps.

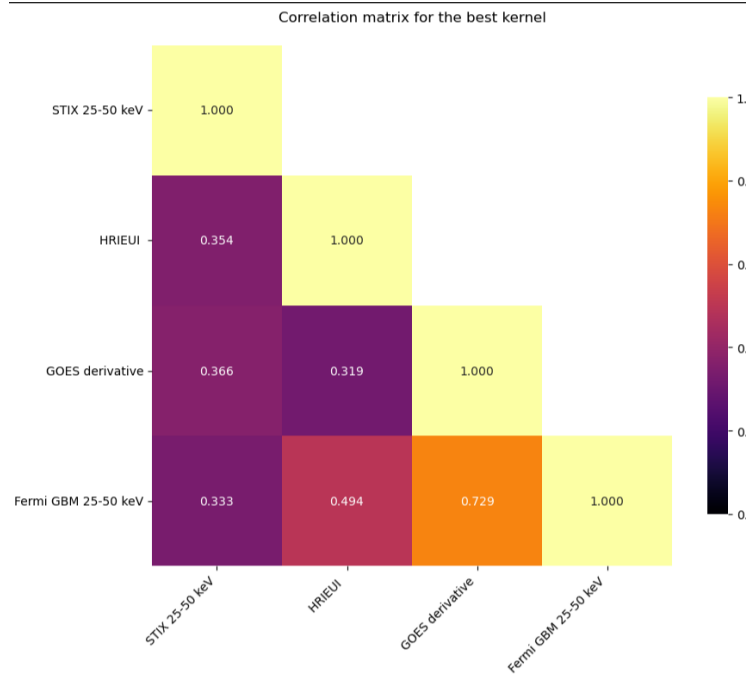
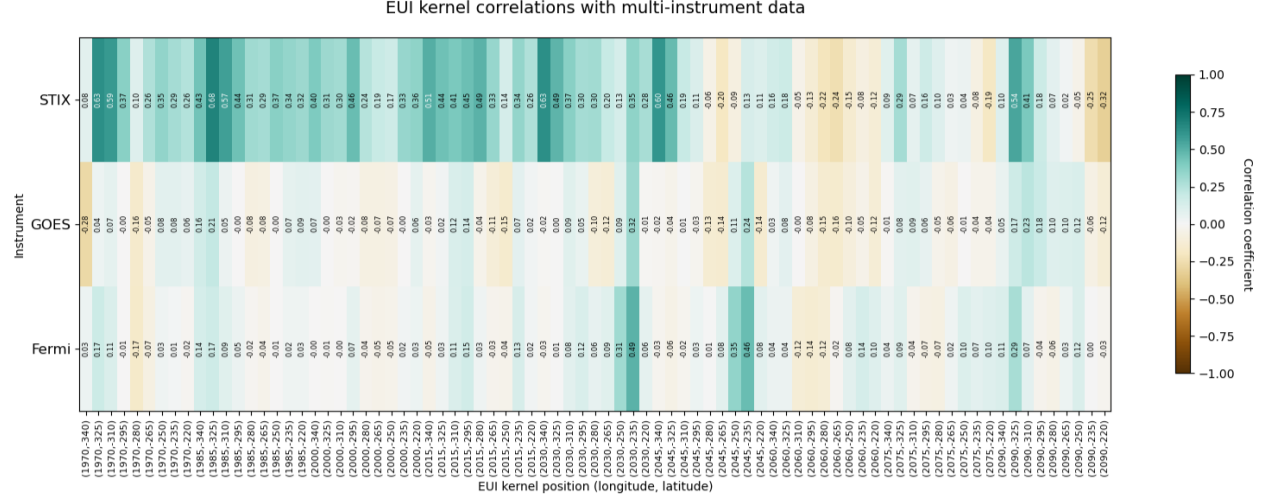


We located one big kernel (black color) and extracted light curve from it. This light curve is not time-corrected (we do that later¹).

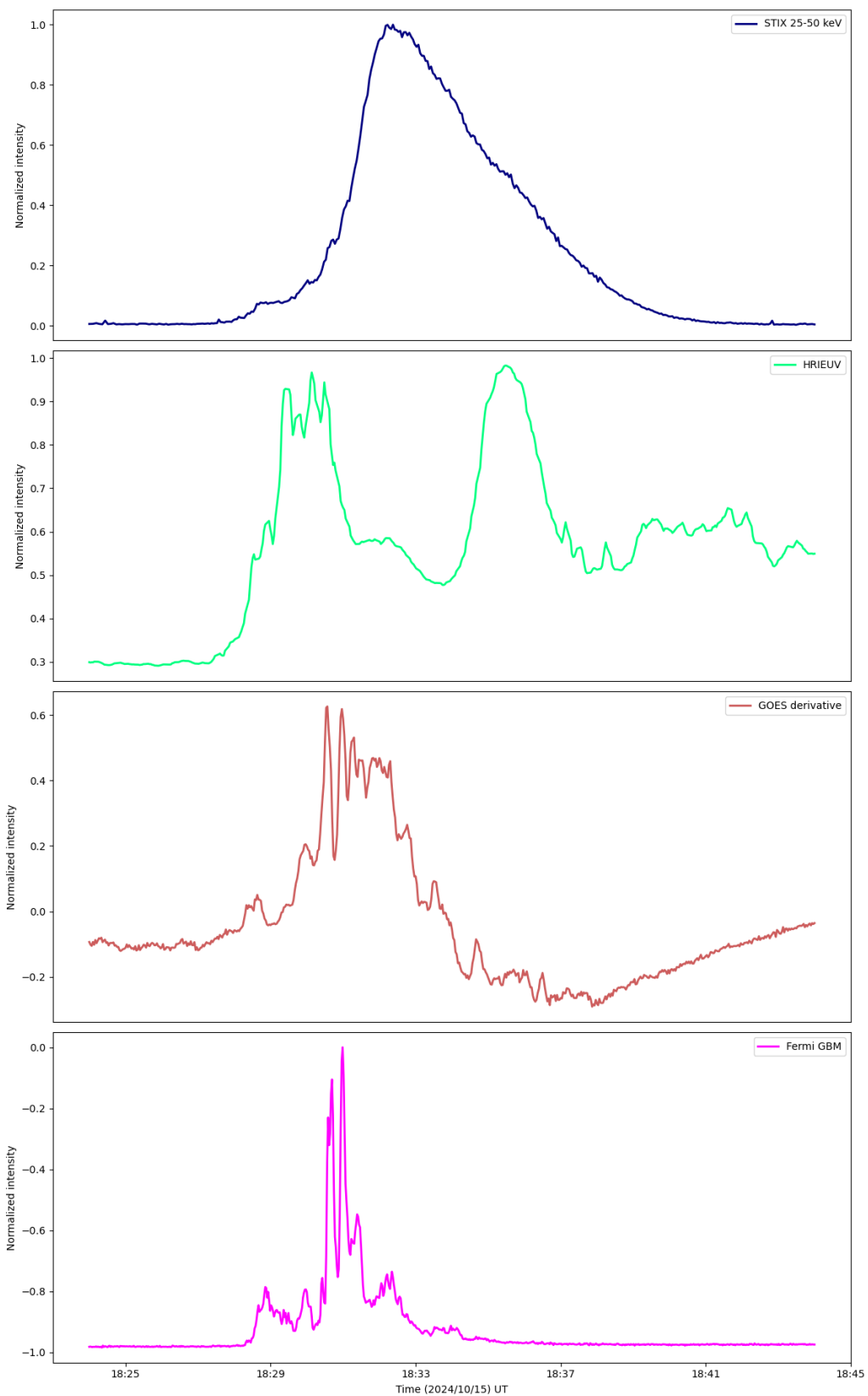


¹There is a time shift between Earth (i.e. GOES/ Fermi) and STIX/EUI of around 218 s.

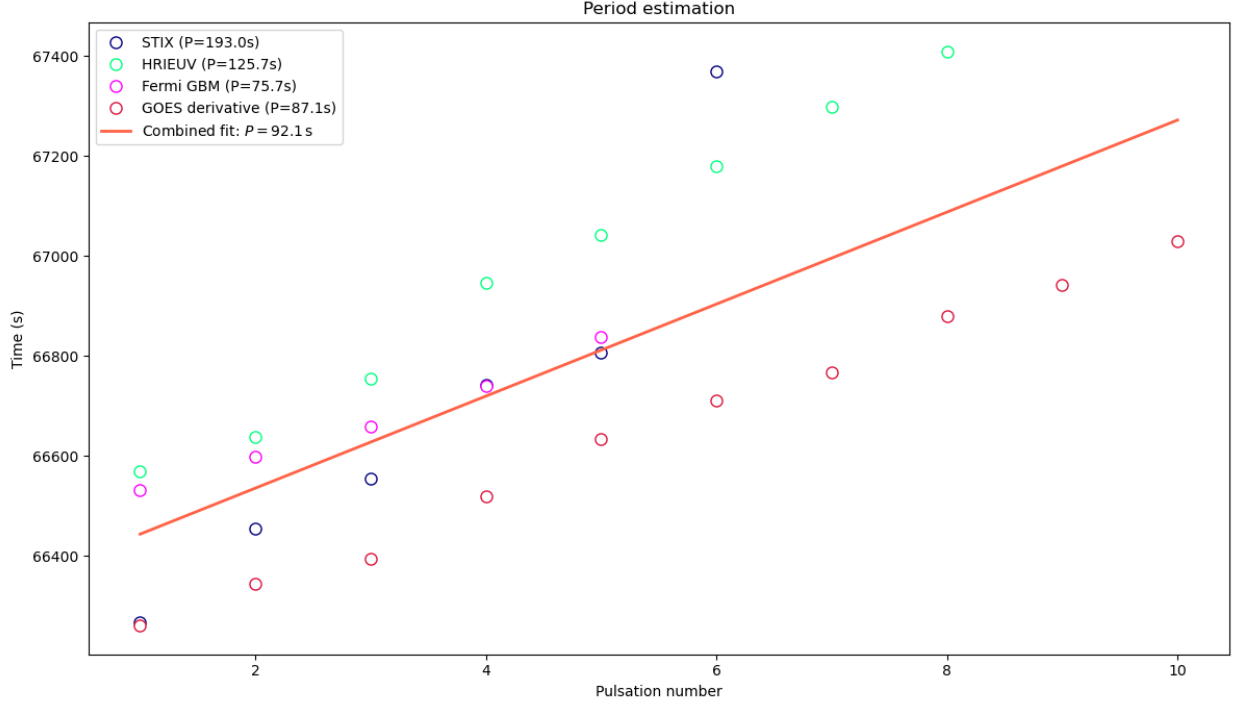
In the similar area to that where black kernel is located, we made multiple (81) smaller kernels ($(30'')^2$), with overlap of $15''$ on all sides. We extracted light curves from all of those and made animation - how light curves are changing shape when different kernel is *sourced*. We can't show animation here. Extension of that part of analysis was the closer expectations of those HRIEUV light curves. We found which one correlates with other light curves (Fermi GBM, STIX, GOES) the most and we used that light curve in determination of periodicities (green one in the resumption).



Light curves from various instruments



Light curves on the previous plots are time-corrected and they're trimmed to the same time range. Periodicities are examined via **wavelet**, **Lomb-Scargle** and **manual** method. In manual method we visually identify peaks, and we only show those results, since we're unsure about first two methods.



Additional remarks

1. We successfully detected quasi-periodic pulsations with a period of 92.1s across four independent instruments (STIX, EU/HRIEUV, Fermi/GBM, GOES). We're not sure about consistency here, and STIX light curve wasn't dealt with in the best way. Period falls within expected range for M-class flares. We need to confirm our findings with Lomb-Scargle and wavelet approach.
2. This period suggests MHD oscillations in the flare arcade (slow-mode waves), and periodic magnetic reconnection modulating particle acceleration. However, we're not quite sure about loop lengths.
3. We need to compare early impulsive vs. peak phase period, because this could reveal changes in the magnetic configuration.

Appendix: period estimation

